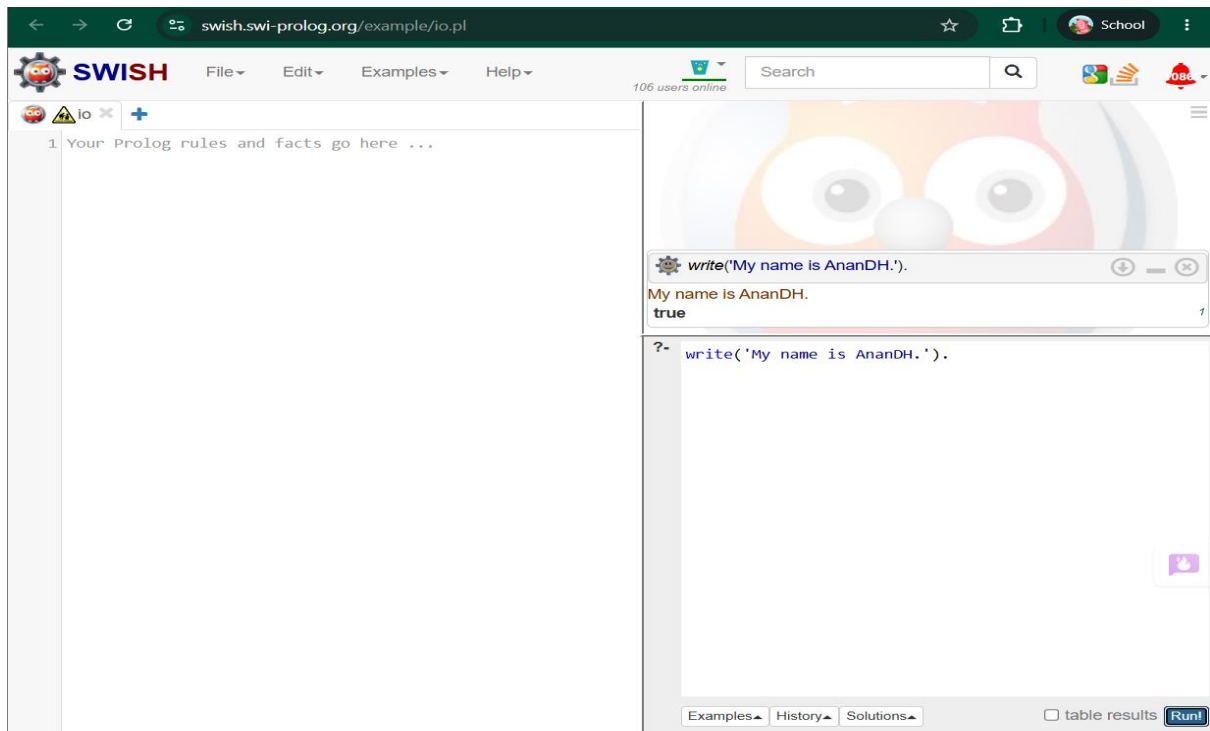


PROLOG PROBLEM

1.



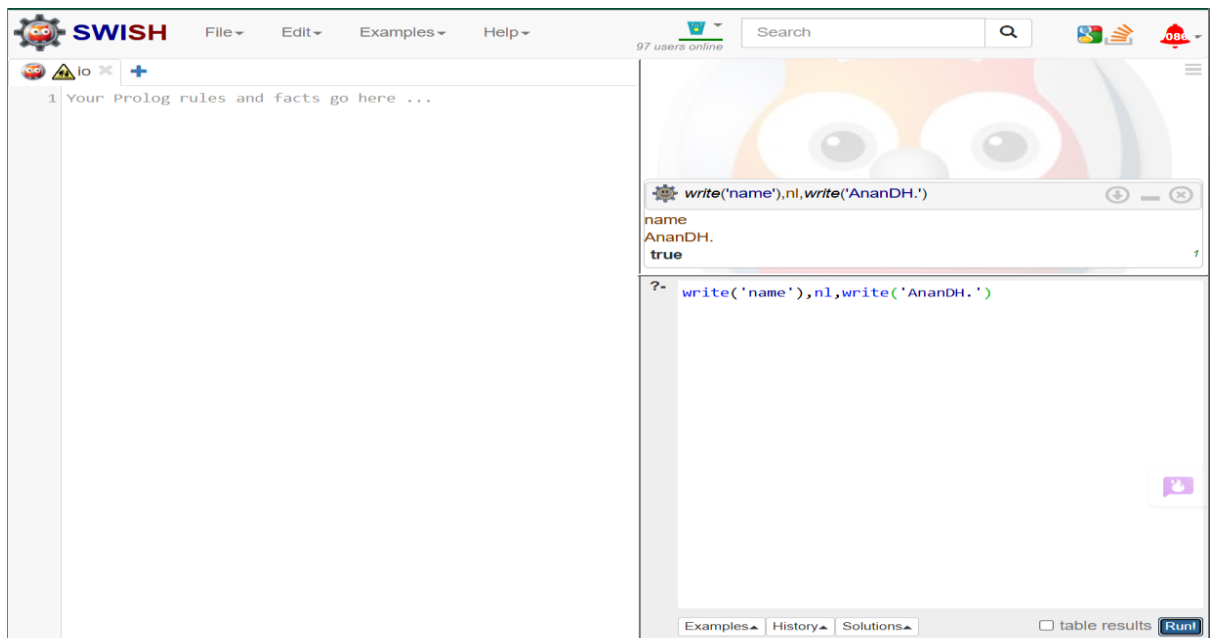
The screenshot shows the SWISH Prolog editor interface. The left pane contains the text "1 Your Prolog rules and facts go here ...". The right pane displays the execution results of a single Prolog query. The query is `write('My name is AnanDH.').`. The output shows the string `My name is AnanDH.` followed by `true`. The interface includes a top menu bar with "File", "Edit", "Examples", and "Help", a search bar, and a status bar indicating "106 users online".

```
1 Your Prolog rules and facts go here ...
```

```
?- write('My name is AnanDH.').
```

My name is AnanDH.
true

2.



The screenshot shows the SWISH Prolog editor interface. The left pane contains the text "1 Your Prolog rules and facts go here ...". The right pane displays the execution results of a Prolog query. The query is `write('name'),nl,write('AnanDH.')`. The output shows the string `name` followed by a newline and the string `AnanDH.`, followed by `true`. The interface includes a top menu bar with "File", "Edit", "Examples", and "Help", a search bar, and a status bar indicating "97 users online".

```
1 Your Prolog rules and facts go here ...
```

```
?- write('name'),nl,write('AnanDH.')
```

name
AnanDH.
true

3.

The screenshot shows the SWISH Prolog IDE interface. The top menu bar includes 'File', 'Edit', 'Examples', and 'Help'. A search bar is located on the right. The left pane contains a single line of text: '1 Your Prolog rules and facts go here ...'. The right pane displays the results of a query. The query is `format("My name is ~nAnanDH.").`. The output shows 'My name is AnanDH.' followed by 'true'. Below the output, there is a section for the query `?- format('My name is ~nAnanDH.').` which is currently empty. At the bottom right, there are buttons for 'Examples', 'History', 'Solutions', a checkbox for 'table results', and a 'Run!' button.

4.

The screenshot shows the SWISH Prolog IDE interface. The top menu bar includes 'File', 'Edit', 'Examples', and 'Help'. A search bar is located on the right. The left pane contains two lines of text: '1 name(anandh).', '2 name(albert).'. The right pane displays the results of a query. The query is `name(X).`. The output shows 'true' followed by two solutions: 'X = anandh' and 'X = albert'. Below the output, there is a section for the query `?- name(X).` which is currently empty. At the bottom right, there are buttons for 'Examples', 'History', 'Solutions', a checkbox for 'table results', and a 'Run!' button.

<https://swish.swi-prolog.org>

5.

The screenshot shows the SWISH web interface. The top navigation bar includes 'File', 'Edit', 'Examples', and 'Help' menus, a search bar, and a status indicator showing '96 users online'. The main editor area contains the following code:

```
1 get_name :-  
2 write('Enter name: '), read(Name), write(Name).
```

The terminal window on the right shows the execution of the 'get_name' function. It displays 'Enter name:' followed by the user input 'dhanu' in a text field. Below the text field, the output 'dhanu' and the return value 'true' are shown. The terminal also displays the prompt '?-' followed by the function name 'get_name'.

6.

The screenshot shows the SWISH web interface. The top navigation bar includes 'File', 'Edit', 'Examples', and 'Help' menus, a search bar, and a status indicator showing '97 users online'. The main editor area contains the following code:

```
1 name :- write('My name is AnanDH.').
```

The terminal window on the right shows the execution of the 'name' function. It displays 'My name is AnanDH.' followed by the return value 'true'. The terminal also displays the prompt '?-' followed by the function name 'name'.

7.

The screenshot shows the SWISH web interface. The top navigation bar includes the SWISH logo, a search bar, and a user count of 101 users online. The left sidebar contains a file explorer with a folder named 'io'. The main editor area displays a code snippet for a function named `get_name`:

```
1 get_name :-  
2 write('Enter your name and age '),  
3 read(Name), read(N),  
4 format('Hello, ~w Your age is ~w', [Name, N]).
```

The right sidebar features a terminal window titled 'get_name'. It shows the execution of the code, with the prompt 'Enter your name and age' and the user input 'dhanu' and '18'. The output is 'Hello, dhanu Your age is 18' and 'true'. Below the terminal is a large text area for comments or additional code, and a 'Run!' button at the bottom right.

8.

The screenshot shows the SWISH web interface. The top navigation bar includes the SWISH logo, a search bar, and a user count of 104 users online. The left sidebar contains a file explorer with a folder named 'io'. The main editor area displays a code snippet for a function named `sum`:

```
1 sum :-  
2 read(A), read(B), C is A + B,  
3 format('Sum = ~w', [C]).
```

The right sidebar features a terminal window titled 'sum.'. It shows the execution of the code, with the prompt 'Sum = ~w' and the user input '5' and '8'. The output is 'Sum = 13' and 'true'. Below the terminal is a large text area for comments or additional code, and a 'Run!' button at the bottom right.

9.

The screenshot shows the SWISH programming environment. The top menu bar includes "File", "Edit", "Examples", and "Help". A search bar is on the right. The left sidebar has a "io" icon and a "+" button. The main editor area contains the following code:

```
1 sum(N,S) :- S is N*(N+1)//2.
```

On the right, there is a large owl illustration. Below it, a window titled "sum(5,S)." shows the variable "S" with the value "15". Below that, a console window titled "?- sum(5,S)." is empty. At the bottom right, there are buttons for "Examples", "History", "Solutions", a checkbox for "table results", and a "Run!" button.

10.

The screenshot shows the SWISH programming environment. The top menu bar includes "File", "Edit", "Examples", and "Help". A search bar is on the right. The left sidebar has a "io" icon and a "+" button. The main editor area contains the following code:

```
1 sum(N,S) :- S is N*(N+1)//2, write(S).
```

On the right, there is a large owl illustration. Below it, a window titled "sum(5,S)." shows the variable "S" with the value "15". Below that, a console window titled "?- sum(5,S)." is empty. At the bottom right, there are buttons for "Examples", "History", "Solutions", a checkbox for "table results", and a "Run!" button.

11.

The screenshot shows the SWISH web interface. The top navigation bar includes the SWISH logo, a search bar, and a user count of 112 users online. The left sidebar contains icons for file management and a plus sign. The main editor area displays the following code:

```
1 si(P,R,T,S) :- S is P*R*T/100.
```

The right sidebar features a large owl illustration. Below it, a console window shows the execution of the function `si(2000,2,2,S)`, resulting in `S = 80`. Below the console, a search bar and a "Run!" button are visible.

12.

The screenshot shows the SWISH web interface. The top navigation bar includes the SWISH logo, a search bar, and a user count of 117 users online. The left sidebar contains icons for file management and a plus sign. The main editor area displays the following code:

```
1 sum(0,0).
2 sum(N,S):-N1 is N-1,sum(N1,S1),S is S1+N.
3 sumofn:-read(N),sum(N,S),write(S).
```

The right sidebar features a large owl illustration. Below it, a console window shows the execution of the function `sumofn`, resulting in the output `15` and `true`. Below the console, a search bar and a "Run!" button are visible.

13.

The SWISH Prolog interface shows a session with the following code in the editor:

```

1 person(ananth, 2004, 10, 25).
2 person(ravi, 2003, 5, 12).
3 person(Diya, 2005, 8, 30).
Singleton variables: [Diya]

```

The right-hand pane displays the results of two queries:

- Query 1:** `person(ananth, Y, M, D).`
 - Singleton variables: [Diya]
 - Results: `D = 25, M = 10, Y = 2004` and `D = 30, M = 8, Y = 2005`.
- Query 2:** `person(Name, 2004, M, D).`
 - Singleton variables: [Diya]
 - Results: `D = 25, M = 10, Name = ananth`.

The bottom of the interface shows buttons for "Examples", "History", "Solutions", a checkbox for "table results", and a "Run!" button.

14.

The SWISH Prolog interface shows a session with the following code in the editor:

```

1 add(A,B,C) :- C is A+B.
2 square(N,S) :- S is N*N.
3 even(N) :- N mod 2 =:= 0.

```

The right-hand pane displays the results of several queries:

- Query 1:** `add(5,6,11).` Result: `true`.
- Query 2:** `square(5,25).` Result: `true`.
- Query 3:** `even(2).` Result: `true`.
- Query 4:** `even(3).` Result: `false`.

The bottom of the interface shows buttons for "Examples", "History", "Solutions", a checkbox for "table results", and a "Run!" button.

15.

SWISH File Edit Examples Help 114 users online Search

```

1 max(A,B,A) :- A>=B.
2 max(A,B,B) :- B>A.

```

max(5,4,5).
true

max(4,5,5).
true

?- **max(4,5,5).**

Examples History Solutions ☐ table results **Run!**

16.

SWISH File Edit Examples Help 116 users online Search

```

1 fact(0,1).
2 fact(N,F) :- N1 is N-1, fact(N1,F1), F is N*F1.

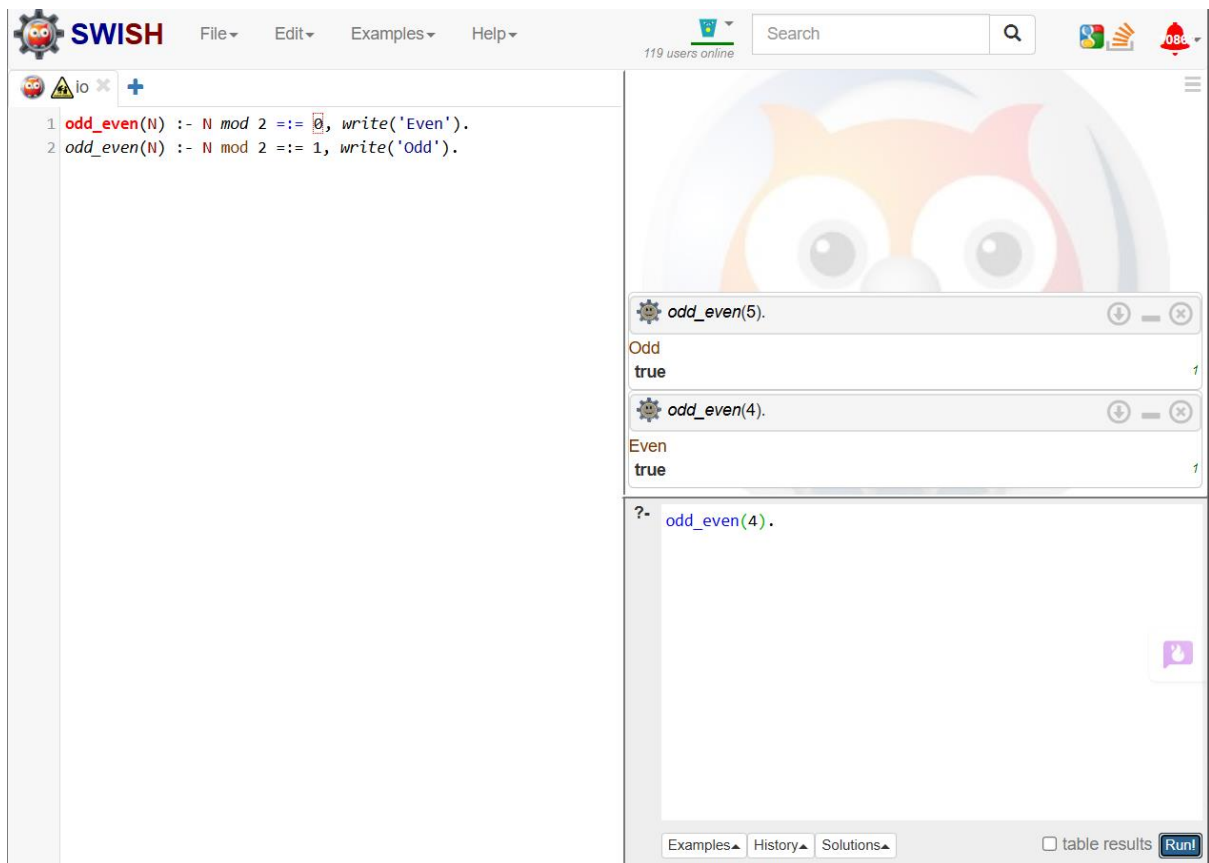
```

fact(0,1).
true

?- **fact(0,1).**

Examples History Solutions ☐ table results **Run!**

17.



The SWISH interface shows a code editor with the following Prolog code:

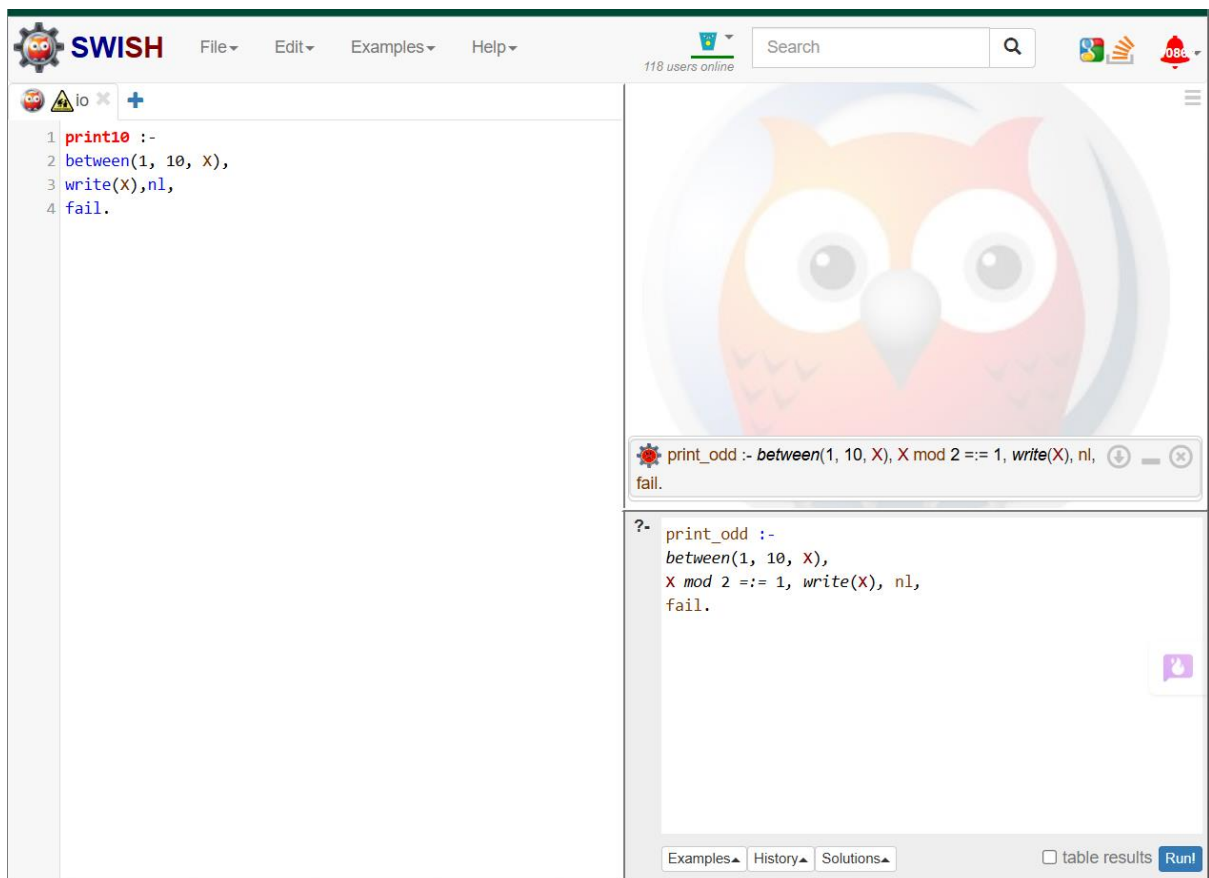
```
1 odd_even(N) :- N mod 2 =:= 0, write('Even').
2 odd_even(N) :- N mod 2 =:= 1, write('Odd').
```

The right-hand side of the interface displays the execution results for two queries:

- Query: `odd_even(5).`
Output: `Odd`
Status: `true`
- Query: `odd_even(4).`
Output: `Even`
Status: `true`

Below the results, there is a search bar with the query `?- odd_even(4).` and a `Run!` button.

18.



The SWISH interface shows a code editor with the following Prolog code:

```
1 print10 :-
2 between(1, 10, X),
3 write(X),nl,
4 fail.
```

The right-hand side of the interface displays the execution results for a query:

- Query: `print_odd :- between(1, 10, X), X mod 2 =:= 1, write(X), nl, fail.`
Output: `fail.`

Below the results, there is a search bar with the query `?- print_odd :- between(1, 10, X), X mod 2 =:= 1, write(X), nl, fail.` and a `Run!` button.

19.

SWISH Pro interface for problem 19. The editor shows the following Prolog code:

```

1 male(john). male(peter). female(mary).
2 parent(john, peter). parent(mary, peter).
3 father(X, Y) :- male(X), parent(X, Y).
4 mother(X, Y) :- female(X), parent(X, Y).

```

The right pane displays the results of the following queries:

- parent(X,Y)**:
 - X = john, Y = peter
 - X = mary, Y = peter
- male(X)**:
 - X = john
 - X = peter
- mother(X,Y)**:
 - X = mary, Y = peter
- mother(mary, peter).**:
 - true

The bottom pane shows the query: `?- mother(mary, peter).`

20.

SWISH Pro interface for problem 20. The editor shows the following Prolog code:

```

1 hanoi(1,A,_C) :- write(A), write(' -> '), write(C), nl.
2 hanoi(N,A,B,C) :-
3   N > 1, M is N-1,
4   hanoi(M,A,C,B), hanoi(1,A,B,C), hanoi(M,B,A,C).

```

The right pane displays the results of the query `hanoi(3,left,middle,right).`:

```

left -> right
left -> middle
right -> middle
left -> right
middle -> left
middle -> right
left -> right
true
false

```

The bottom pane shows the query: `?- hanoi(3,left,middle,right).`